Chapter 28

RSA Accelerator (RSA)

28.1 Introduction

The RSA accelerator provides hardware support for high-precision computation used in various RSA asymmetric cipher algorithms, significantly improving their run time and reducing their software complexity compared with RSA algorithms implemented solely in software. The RSA accelerator also supports operands of different lengths, which provides more flexibility during the computation.

28.2 Features

The following functionality is supported:

- Large-number modular exponentiation with two additional acceleration options to further increase the calculation speed
- Large-number modular multiplication
- Large-number multiplication
- Operands of different lengths
- Interrupt on completion of computation

28.3 Functional Description

The RSA accelerator is activated by setting the HP_SYS_CLKRST_REG_CRYPTO_RSA_CLK_EN bit in the HP_SYS_CLKRST_PERI_CLK_CTRL25_REG register and clearing the HP_SYS_CLKRST_REG_RST_EN_RSA bit in the HP_SYS_CLKRST_HP_RST_EN2_REG register. Additionally, users also need to clear HP_SYS_CLKRST_REG_RST_EN_DS and HP_SYS_CLKRST_REG_RST_EN_ECDSA bits to reset the RSA Digital Signature Peripheral (RSA_DS) and the ECDSA Digital Signature Peripheral (ECDSA_DS) accelerators.

The RSA accelerator is only available after the RSA-related memories are initialized. The content of the RSA_QUERY_CLEAN_REG register is 0 during initialization and will become 1 after the initialization is done. Therefore, wait until RSA_QUERY_CLEAN_REG becomes 1 before using the RSA accelerator.

The RSA_INT_ENA_REG register is used to control the interrupt triggered on completion of computation. Write 1 or 0 to this field to enable or disable the interrupt. By default, the interrupt function of the RSA accelerator is enabled.

Notice:

ESP32-P4's RSA Digital Signature Peripheral (RSA_DS) and ECDSA Digital Signature Peripheral (ECDSA_DS) modules

also call the RSA accelerator when working. Therefore, users cannot access the RSA accelerator when the RSA Digital Signature Peripheral (RSA_DS) or the ECDSA Digital Signature Peripheral (ECDSA_DS) module is working.

28.3.1 Definitions and Representations

ESP32-P4's RSA accelerator supports operands of different lengths $N = 32 \times n$.

To represent the numbers used as operands, let us define a base-b positional notation, as follows:

$$b = 2^{32}$$

Using this notation, each number is represented by a sequence of base-b digits:

$$n = \frac{N}{32}$$
$$Z = (Z_{n-1}Z_{n-2}\cdots Z_0)_b$$
$$X = (X_{n-1}X_{n-2}\cdots X_0)_b$$
$$Y = (Y_{n-1}Y_{n-2}\cdots Y_0)_b$$
$$M = (M_{n-1}M_{n-2}\cdots M_0)_b$$
$$\overline{r} = (\overline{r}_{n-1}\overline{r}_{n-2}\cdots \overline{r}_0)_b$$

Each of the values in $Z_{n-1}\cdots Z_0$, $X_{n-1}\cdots X_0$, $Y_{n-1}\cdots Y_0$, $M_{n-1}\cdots M_0$, $\overline{r}_{n-1}\cdots \overline{r}_0$ represents one base-b digit (a 32-bit word).

Z_{n-1}, X_{n-1}, Y_{n-1}, M_{n-1} and $\overline{r}_{n-1}$ are the most significant bits of Z, X, Y, M, while Z_0, X_0, Y_0, M_0 and $\overline{r}_0$ are the least significant bits.

If we define $R = b^n$, the additional argument $\overline{r}$ can be calculated as $\overline{r} = R^2 \bmod M$.

Also, argument M' can be calculated using the formula below:

$$M' = -M^{-1} \bmod b$$

where, M^{-1} is the modular multiplicative inverse of M, and it can be calculated with the extended binary GCD algorithm.

28.3.2 Large-Number Modular Exponentiation

Large-number modular exponentiation performs $Z = X^Y \bmod M$. The computation is based on Montgomery multiplication. Therefore, aside from the X, Y, and M arguments, two additional ones are needed — $\overline{r}$ and M', which need to be calculated in advance by software.

The RSA accelerator supports operands of length $N = 32 \times n$, where $n \in \{1, 2, 3, \ldots, 128\}$. The bit lengths of arguments Z, X, Y, M, and $\overline{r}$ can be arbitrary N, but all numbers in a calculation must be of the same length. The bit length of M' must be 32.

Large-number modular exponentiation on the ESP32-P4 can be implemented as follows:

1. Write 1 or 0 to the RSA_INT_ENA field to enable or disable the interrupt function.
2. Configure relevant registers:
   (a) Write $(\frac{N}{32} - 1)$ to the RSA_MODE_REG register.

(b) Write M' to the RSA_M_PRIME_REG register.

(c) Configure registers related to the acceleration options, which are described later in Section 28.3.5.

3. Write X_i, Y_i, M_i and $\overline{r}_i$ for $i \in \{0, 1, \ldots, n-1\}$ to memory blocks RSA_X_MEM, RSA_Y_MEM, RSA_M_MEM and RSA_Z_MEM. The capacity of each memory block is 128 words. Each word of each memory block can store one base-b digit. The memory blocks use the little endian format for storage, i.e., the least significant digit of each number is in the lowest address.

   Users need to write data to each memory block only according to the length of the number; data beyond this length is ignored.

4. Write 1 to the RSA_SET_START_MODEXP field of the RSA_SET_START_MODEXP_REG register to start computation.
5. Wait for the completion of computation, which happens when the content of RSA_QUERY_IDLE becomes 1 or the RSA interrupt occurs, if enabled.
6. Read the result Z_i for $i \in \{0, 1, \ldots, n-1\}$ from RSA_Z_MEM.
7. If you have the interrupt enabled, write 1 to RSA_CLEAR_INTERRUPT to clear the interrupt.

After the computation, the RSA_MODE_REG register, memory blocks RSA_Y_MEM and RSA_M_MEM, as well as the RSA_M_PRIME_REG remain unchanged. However, X_i in RSA_X_MEM and $\overline{r}_i$ in RSA_Z_MEM computation are overwritten, and only these overwritten memory blocks need to be re-initialized before starting another computation.

28.3.3 Large-Number Modular Multiplication

Large-number modular multiplication performs $Z = X \times Y \bmod M$. This computation is based on Montgomery multiplication. Therefore, similar to the large-number modular exponentiation, two additional arguments are needed – $\overline{r}$ and M', which need to be calculated in advance by software.

The RSA accelerator supports large-number modular multiplication with operands of length $N = 32 \times n$, where $n \in \{1, 2, 3, \ldots, 128\}$.

The computation can be executed as follows:

1. Write 1 or 0 to the RSA_INT_ENA_REG register to enable or disable the interrupt function.
2. Configure relevant configuration registers:

   (a) Write $(\frac{N}{32} - 1)$ to the RSA_MODE_REG register.

   (b) Write M' to the RSA_M_PRIME_REG register.

3. Write X_i, Y_i, M_i, and $\overline{r}_i$ for $i \in \{0, 1, \ldots, n-1\}$ to memory blocks RSA_X_MEM, RSA_Y_MEM, RSA_M_MEM, and RSA_Z_MEM, respectively. The capacity of each memory block is 128 words. Each word of each memory block can store one base-b digit. The memory blocks use the little endian format for storage, i.e., the least significant digit of each number is in the lowest address.

   Users need to write data to each memory block only according to the length of the number; data beyond this length are ignored.

4. Write 1 to the RSA_SET_START_MODMULT field.

5. Wait for the completion of computation, which happens when the content of RSA_QUERY_IDLE becomes 1 or the RSA interrupt occurs, if enabled.
6. Read the result Z_i for $i \in \{0, 1, \ldots, n-1\}$ from RSA_Z_MEM.
7. If you have the interrupt enabled, write 1 to RSA_CLEAR_INTERRUPT to clear the interrupt, .

After the computation, the length of operands in RSA_MODE_REG, the X_i in memory RSA_X_MEM, the Y_i in memory RSA_Y_MEM, the M_i in memory RSA_M_MEM, and the M' in memory RSA_M_PRIME_REG remain unchanged. However, the $\overline{r}_i$ in memory RSA_Z_MEM has already been overwritten, and only this overwritten memory block needs to be re-initialized before starting another computation.

28.3.4 Large-Number Multiplication

Large-number multiplication performs $Z = X \times Y$. The length of result Z is twice that of operand X and operand Y. Therefore, the RSA accelerator only supports large-number multiplication with operand length $N = 32 \times n$, where $n \in \{1, 2, 3, \ldots, 64\}$. The length $\hat{N}$ of result Z is $2 \times N$.

The computation can be executed as follows:

1. Write 1 or 0 to the RSA_INT_ENA_REG register to enable or disable the interrupt function.
2. Write $(\frac{\hat{N}}{32} - 1)$, i.e., $(\frac{N}{16} - 1)$ to the RSA_MODE_REG register.
3. Write X_i and Y_i for $\in \{0, 1, \ldots, n-1\}$ to memory blocks RSA_X_MEM and RSA_Z_MEM. Each word of each memory block can store one base-b digit. The memory blocks use the little endian format for storage, i.e., the least significant digit of each number is in the lowest address. n is $\frac{N}{32}$.

   Write X_i for $i \in \{0, 1, \ldots, n-1\}$ to the address of the i words of the RSA_X_MEM memory block. Note that Y_i for $i \in \{0, 1, \ldots, n-1\}$ will not be written to the address of the i words of the RSA_Z_MEM register, but the address of the $n + i$ words, i.e., the base address of the RSA_Z_MEM memory plus the address offset $4 \times (n + i)$.

   Users need to write data to each memory block only according to the length of the number; data beyond this length is ignored.
4. Write 1 to the RSA_SET_START_MULT register.
5. Wait for the completion of computation, which happens when the content of RSA_QUERY_IDLE becomes 1 or the RSA interrupt occurs, if enabled.
6. Read the result Z_i for $i \in \{0, 1, \ldots, \hat{n}-1\}$ from the RSA_Z_MEM register. $\hat{n}$ is $2 \times n$.
7. If you have the interrupt enabled, write 1 to RSA_CLEAR_INTERRUPT to clear the interrupt.

After the computation, the length of operands in RSA_MODE_REG and the X_i in memory RSA_X_MEM remain unchanged. However, the Y_i in memory RSA_Z_MEM has already been overwritten, and only this overwritten memory block needs to be re-initialized before starting another computation.

28.3.5 Options for Additional Acceleration

The ESP32-P4 RSA accelerator also provides SEARCH and CONSTANT_TIME options that can be configured to further accelerate the large-number modular exponentiation. By default, both options are configured as no additional acceleration.

Users can choose to use one or two of these options to further accelerate the computation. Note that, even when none of these two options is configured, using the hardware RSA accelerator is still much faster than implementing the RSA algorithm in software.

To be more specific, when neither of these two options are configured for additional acceleration, the time required to calculate $Z = X^Y \bmod M$ is solely determined by the lengths of operands. When either or both of these two options are configured for additional acceleration, the time required is also correlated with the 0 and 1 bit distribution in Y.

To better illustrate how these two options work, first assume Y is represented in binaries as

$$Y = (\widetilde{Y}_{N-1}\widetilde{Y}_{N-2}\cdots\widetilde{Y}_{t+1}\widetilde{Y}_t\widetilde{Y}_{t-1}\cdots\widetilde{Y}_0)_2$$

where,

- N is the length of Y,
- $\widetilde{Y}_t$ is 1,
- $\widetilde{Y}_{N-1}$, $\widetilde{Y}_{N-2}$, ..., $\widetilde{Y}_{t+1}$ are all equal to 0,
- and $\widetilde{Y}_{t-1}$, $\widetilde{Y}_{t-2}$, ..., $\widetilde{Y}_0$ are either 0 or 1 but exactly m bits should be equal to 0 and t-m bits 1, i.e., the Hamming weight of $\widetilde{Y}_{t-1}\widetilde{Y}_{t-2}, \cdots, \widetilde{Y}_0$ is $t - m$.

When either of these two options is configured for additional acceleration:

- SEARCH Option (Configuring RSA_SEARCH_ENABLE to 1 for additional acceleration)
  - The accelerator ignores the bit positions of $\widetilde{Y}_i$, where $i > \alpha$. Search position α is set by configuring the RSA_SEARCH_POS_REG register. Set α to a number smaller than N-1, which otherwise leads to the same result as if this option is not used for additional acceleration. The best acceleration performance can be achieved by setting α to t, in which case all the $\widetilde{Y}_{N-1}$, $\widetilde{Y}_{N-2}$, ..., $\widetilde{Y}_{t+1}$ of 0 s are ignored during the calculation. Note that if you set α to be less than t, then the result of the modular exponentiation $Z = X^Y \bmod M$ will be incorrect.
  - Note that this option compromises the security because it ignores some bits, which essentially shortens the key length, thus should not be enabled for applications with high security requirement.
- CONSTANT_TIME Option (Configuring RSA_CONSTANT_TIME_REG to 0 for additional acceleration)
  - The accelerator speeds up the calculation by simplifying the calculation concerning the 0 bits of Y. Therefore, the higher the proportion of bits 0 against bits 1, the better is the acceleration performance.
  - Note that this option also compromises the security because its time cost correlates with the 0/1 distribution of the key, which can be used in a Side Channel Attack (SCA), thus should not be enabled for applications with high security requirement.

Below is an example to demonstrate the performance of the RSA accelerator under different combinations of SEARCH and CONSTANT_TIME configuration. In this example:

- We perform $Z = X^Y \bmod M$
- N = 3072

- Y = 65537
- X and M are taken at random
- When the SEARCH option is enabled, α, i.e., the position register RSA_SEARCH_POS_REG, is set to 16.

Table 28.3-1 below demonstrates the time cost in clock cycles under different combinations of SEARCH and CONSTANT_TIME configuration when performing $Z = X^Y \bmod M$ as described above.

Table 28.3-1. Acceleration Performance

SEARCH Option	CONSTANT_TIME Option	Time Cost (clock cycle)
No acceleration	No acceleration	174.7×10^6
Accelerated	No acceleration	1.023×10^6
No acceleration	Acceleration	0.546×10^6
Acceleration	Acceleration	0.540×10^6

As shown in Table 28.3-1:

- The time cost is biggest when none of these two options is configured for additional acceleration.
- The time cost is smallest when both of these two options are configured for additional acceleration.
- The time cost can be dramatically reduced when either or both option(s) are configured for additional acceleration.

28.4 Interrupts

ESP32-P4's RSA accelerator can generate the following interrupt signal that will be sent to the Interrupt Matrix.

- RSA_INTR

There is one internal interrupt source from the RSA accelerator that can generate the above interrupt signal. The interrupt source from the RSA accelerator is listed with its trigger condition and the resulting interrupt signal in Table 28.4-1.

Table 28.4-1. RSA's Internal Interrupt Source

Internal Interrupt Source	Trigger Condition	Interrupt Signal
RSA_CALC_DONE_INT	Completion of an RSA calculation	RSA_INTR

Note:
For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter 12 *Interrupt Matrix* > Section 12.2 *Interrupt Terminology in ESP32-P4*.

28.5 Memory Summary

The addresses in this section are relative to the RSA accelerator base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Table 28.5-1. RSA Accelerator Memory Blocks

Name	Description	Size (byte)	Starting Address	Ending Address	Access
RSA_M_MEM	Memory M	512	0x0000	0x01FF	R/W
RSA_Z_MEM	Memory Z	512	0x0200	0x03FF	R/W
RSA_Y_MEM	Memory Y	512	0x0400	0x05FF	R/W
RSA_X_MEM	Memory X	512	0x0600	0x07FF	R/W

28.6 Register Summary

The addresses in this section are relative to the RSA accelerator base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

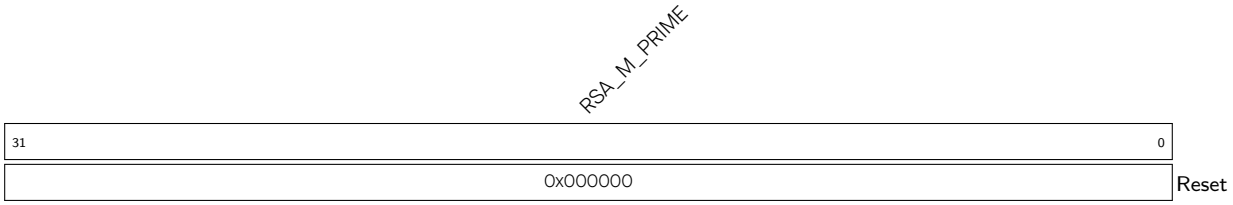
Name	Description	Address	Access
Control or Configuration Registers			
RSA_M_PRIME_REG	Represents M'	0x0800	R/W
RSA_MODE_REG	Configures RSA length	0x0804	R/W
RSA_SET_START_MODEXP_REG	Starts modular exponentiation	0x080C	WT
RSA_SET_START_MODMULT_REG	Starts modular multiplication	0x0810	WT
RSA_SET_START_MULT_REG	Starts multiplication	0x0814	WT
RSA_QUERY_IDLE_REG	Represents the RSA status	0x0818	RO
RSA_CONSTANT_TIME_REG	Configures the constant_time option	0x0820	R/W
RSA_SEARCH_ENABLE_REG	Configures the search option	0x0824	R/W
RSA_SEARCH_POS_REG	Configures the search position	0x0828	R/W
Status Register			
RSA_QUERY_CLEAN_REG	RSA initialization status	0x0808	RO
Interrupt Registers			
RSA_INT_CLR_REG	Clears RSA interrupt	0x081C	WT
RSA_INT_ENA_REG	Enables the RSA interrupt	0x082C	R/W
Version Control Register			
RSA_DATE_REG	Version control register	0x0830	R/W

28.7 Registers

The addresses in this section are relative to the RSA accelerator base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

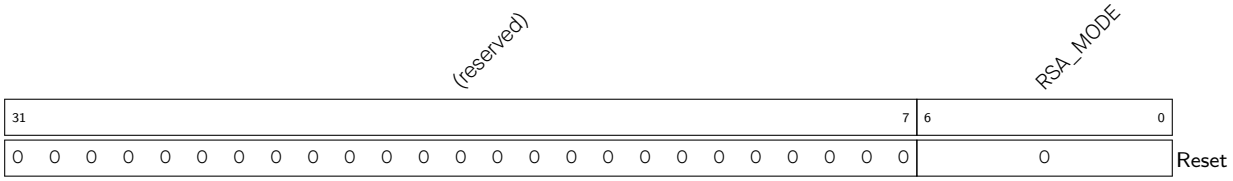
For how to program reserved fields, please refer to Section IX .

Register 28.1. RSA_M_PRIME_REG (0x0800)



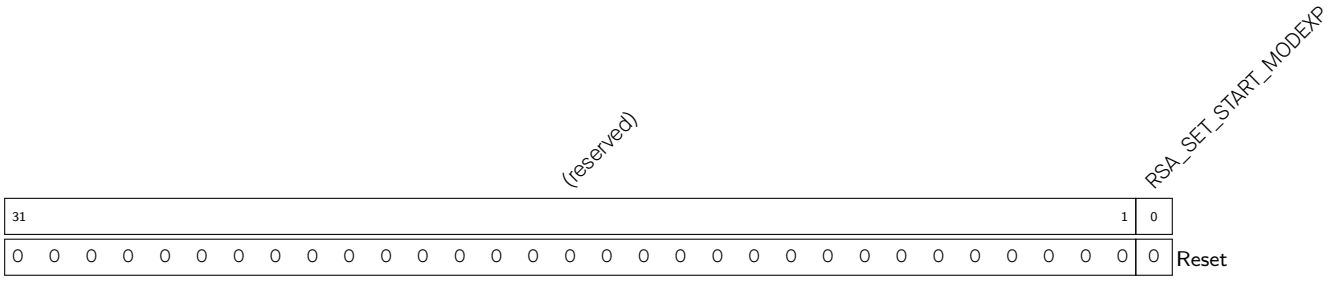
RSA_M_PRIME Represents M'. (R/W)

Register 28.2. RSA_MODE_REG (0x0804)



RSA_MODE Configures the RSA length. (R/W)

Register 28.3. RSA_SET_START_MODEXP_REG (0x080C)



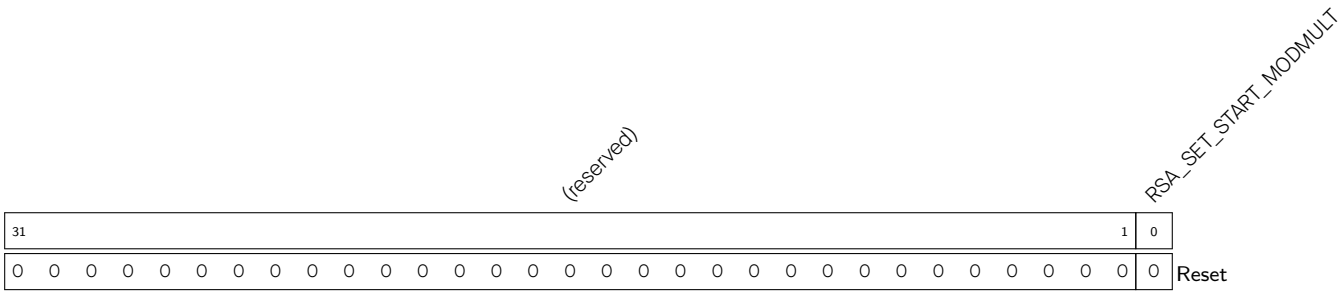
RSA_SET_START_MODEXP Configures whether or not to starts the modular exponentiation.

0: No effect

1: Start

(WT)

Register 28.4. RSA_SET_START_MODMULT_REG (0x0810)



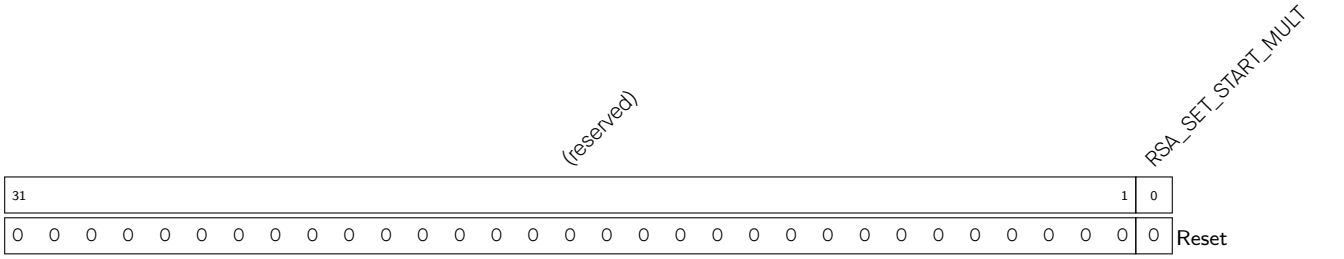
RSA_SET_START_MODMULT Configures whether or not to start the modular multiplication.

0: No effect

1: Start

(WT)

Register 28.5. RSA_SET_START_MULT_REG (0x0814)



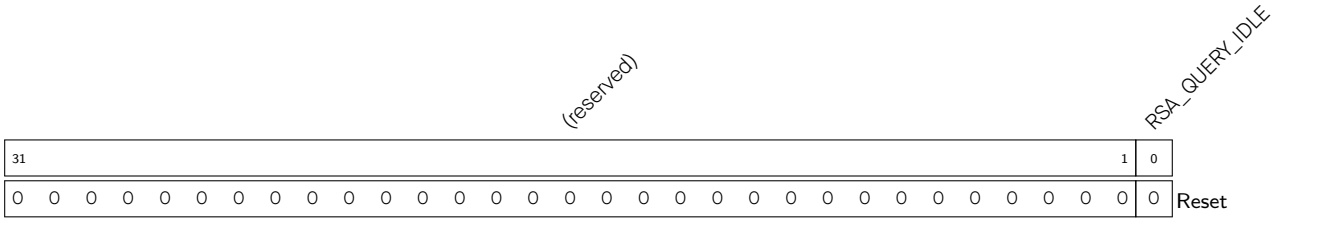
RSA_SET_START_MULT Configures whether or not to start the multiplication.

0: No effect

1: Start

(WT)

Register 28.6. RSA_QUERY_IDLE_REG (0x0818)



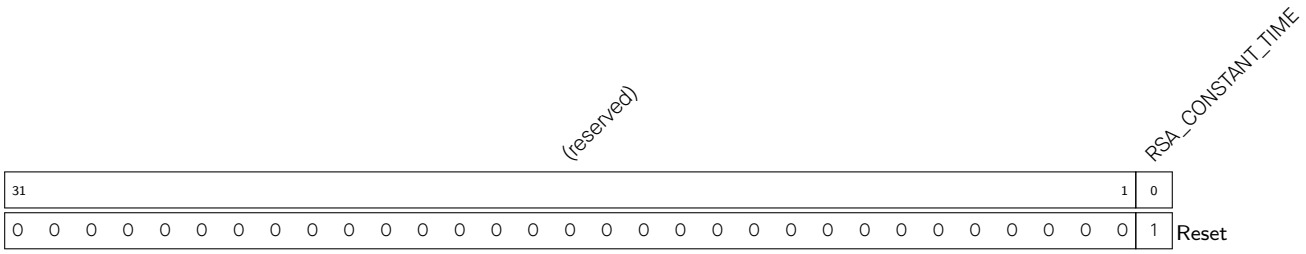
RSA_QUERY_IDLE Represents the RSA status.

0: Busy

1: Idle

(RO)

Register 28.7. RSA_CONSTANT_TIME_REG (0x0820)



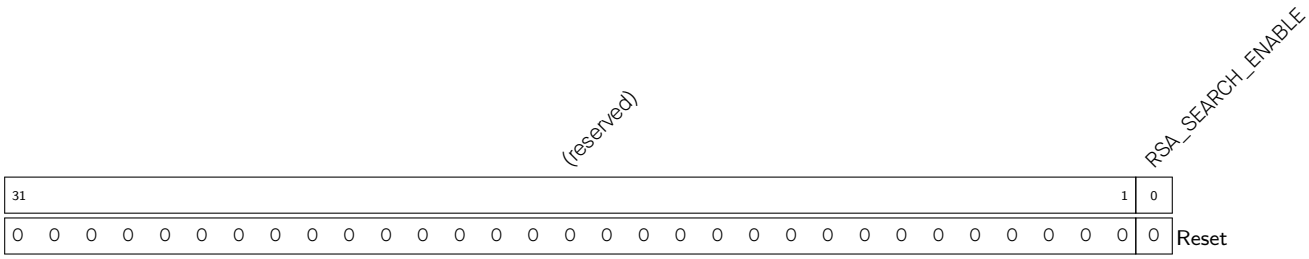
RSA_CONSTANT_TIME Configures the constant_time option.

0: Acceleration

1: No acceleration (default)

(R/W)

Register 28.8. RSA_SEARCH_ENABLE_REG (0x0824)



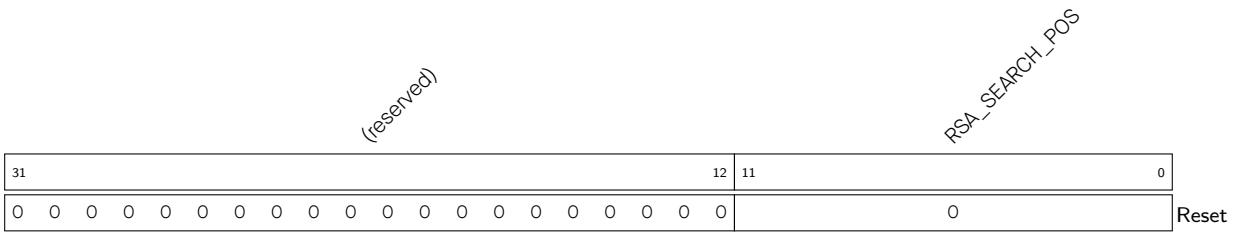
RSA_SEARCH_ENABLE Configures the search option.

0: No acceleration (default)

1: Acceleration

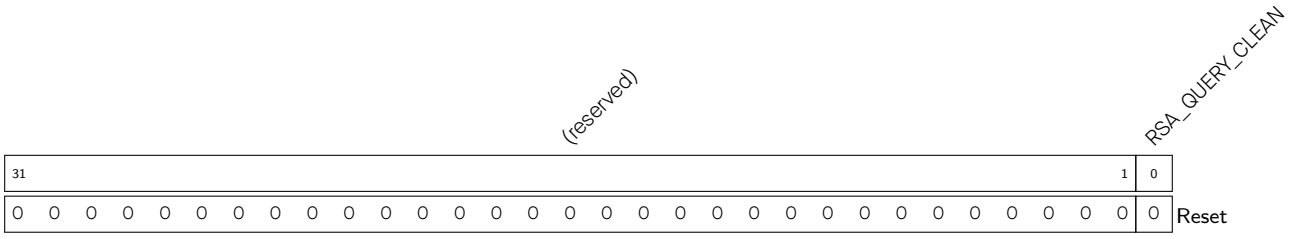
This option should be used together with RSA_SEARCH_POS_REG. (R/W)

Register 28.9. RSA_SEARCH_POS_REG (0x0828)



RSA_SEARCH_POS Configures the starting address to start search. This field should be used together with RSA_SEARCH_ENABLE_REG. The field is only valid when RSA_SEARCH_ENABLE is high. (R/W)

Register 28.10. RSA_QUERY_CLEAN_REG (0x0808)



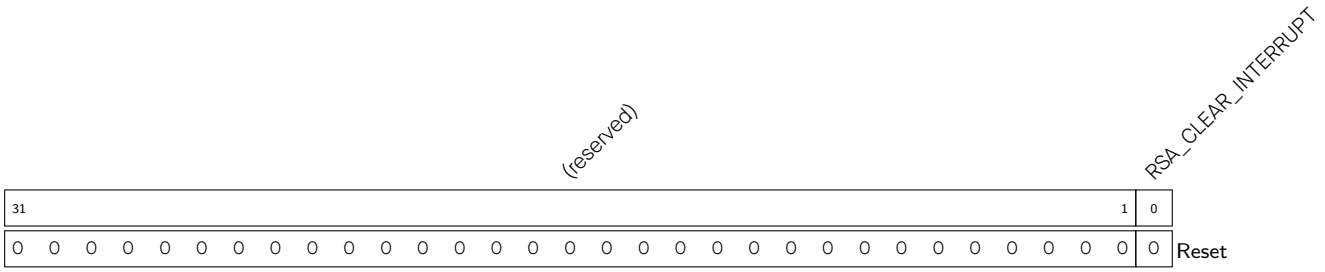
RSA_QUERY_CLEAN Represents whether or not the RSA memory completes initialization.

0: Not complete

1: Completed

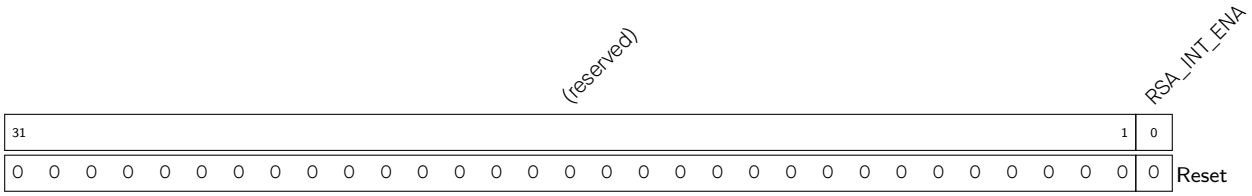
(RO)

Register 28.11. RSA_INT_CLR_REG (0x081C)



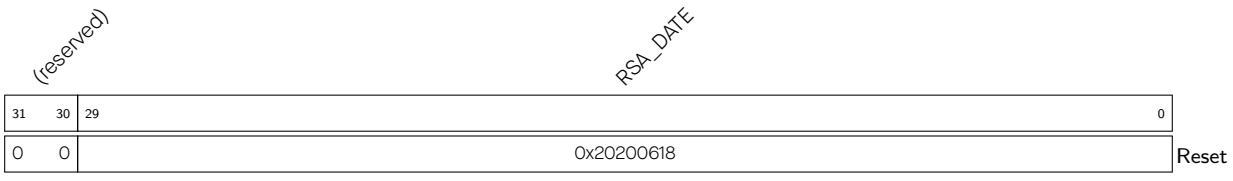
RSA_CLEAR_INTERRUPT Write 1 to clear the RSA interrupt. (WT)

Register 28.12. RSA_INT_ENA_REG (0x082C)



RSA_INT_ENA Write 1 to enable the RSA interrupt. (R/W)

Register 28.13. RSA_DATE_REG (0x0830)



RSA_DATE Version control register. (R/W)